LaMont D. Walker

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Education

Ph.D., Mathematics, Howard University, Washington, DC, expected June 2030

B.A., Mathematics, University of Washington, Seattle, WA, June 2026

Relevant Coursework

Advanced Linear Algebra, Differential Geometry, Modern Algebra, Topology, Complex Analysis, Real Analysis

Research Experience

Summer Research in Mathematics at UW

Seattle, WA, Summer 2025

- Gave lecture on Category Theory and Operadic Lie Algebras
- Read Hu's and Venkatesh's papers on Ver4+
- Learned about monoidal categories

Undergraduate Research Fellowship at UW

Seattle, WA, Summer 2023

- Programmed simulation for the molecular orbitals of a polyacetylene chain using the SSH model
- Applied quantum mechanics and topology to understand the properties of topological insulators
- Gave lectures on the expressions of Maxwell's equations in reciprocal space

Independent Reading

Foundations of Algebraic Geometry, Vakil

- Translate between vanishing sets, their associated ideals and the distinguished open sets of the spectrum of a commutative ring.
- Prove the spectrum is compact using a partition of unity argument
- Show that a morphism of ringed spaces induces a homomorphism of the associated stalks

- Prove the lying over theorem by showing that for any field which is an integral extension of one of its subrings, that subring is also a field.

Galois Theory, Leinster

- View the Galois group of a polynomial as the subgroup of the permutations which fix conjugate tuples of the roots.
- Learn about the Galois correspondence, the connection with solvability of groups to solvability of a polynomial by radicals, how to use the Galois correspondence to classify finite fields.

Algebraic Topology, Hatcher

- Learn about deformation retractions, homotopy, homotopy equivalence, the fundamental group and the Van Kampen theorem
- Learn the proof of Brouwer's Fixed Point Theorem on a closed disk using the fundamental group of the circle.

Category Theory in Context, Riehl

- Learn about functors, natural transformations, limits and colimits, adjunctions and Kan extensions.
- Derive limits, colimits, and adjunctions as special cases of Kan extensions
- Prove the Yoneda Lemma

Expository Writing

On the Classification of Finite Groups with Maximal Subgroups of Prime Order, 2025,

https://llwll2026.github.io/On%20the%20Classification%20of%20Finite%20Groups%20with%20Maximal%20Subgroups%20of%20Prime%20Order.pdf

Awards

Early entrance into the University of Washington through the Robinson Center

Rising Chem Star Undergraduate Scholarship

John and Kathy Connors Foundation Endowed Scholarship

Research Interests

Hilbert schemes of nonreduced algebraic curves

Modular forms, elliptic curves and Galois representations

Combinatorial group theory